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Existence and Uniqueness of Solutions of a General Type Boundary Value Problem for Second Order Functional Differential Equations

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1. INTRODUCTION

In a recent paper, Pachpatte [10] studied a boundary value problem (BVP for short) that consists of an integrodifferential equation of the form

$$x''(t) = f\left(t, x(t), x'(t), \int_a^t h(t, s, x(s), x'(s))\, ds\right), \qquad t \in [a, b]$$

and of the boundary conditions

$$x(a) = A, \qquad x(b) = B.$$

In this problem $f: [a, b] \times (\mathbb{R}^n)^3 \to \mathbb{R}^n$ and $h: [a, b]^2 \times (\mathbb{R}^n)^2 \to \mathbb{R}^n$ are supposed continuous functions and A, B are constants in $\mathbb{R}^n$. Under appropriate Lipschitz conditions on f and h, in [10], it is proved that the above BVP has a unique solution.

Also, in [6] the authors have established existence and comparison results for a class of more general functional differential systems.

In this paper we shall establish existence and uniqueness results for a BVP for a class of functional differential equations like those considered in [6]. Thus, the results of our paper generalize previous ones (e.g., [5, 8, 9]) which concern ordinary or functional differential equations, and also include as special cases results for BVP for integrodifferential equations, like those mentioned in the beginning.

Our paper is organized as follows. In Section 2 we present some preliminaries and state some well-known lemmas together with the formulation of the BVP which we study here. The existence and uniqueness

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of solutions is discussed in Section 3, where some results on a priori bounds of solutions are also given. Finally, in Section 4 we obtain a result on continuous dependence.

2. PRELIMINARIES

Let $\mathbb{R}$ be the set of real numbers, $\mathbb{R}^n$ the n-dimensional Euclidean space and $|\cdot|$ the euclidean norm in $\mathbb{R}^n$. Also, if a, b are real numbers and $x: [a, b] \to \mathbb{R}^n$ is a continuous function we set

$$\|x\|_{[a,b]} = \sup\{|x(t)|: t \in [a, b]\}.$$

Moreover, let r be a positive real number and C_r be the real Banach space of all continuous functions $x: [-r, 0] \to \mathbb{R}^n$ endowed with the sup norm $\|\cdot\|_{[-r,0]}$.

We suppose that T is a positive number and for any continuous function $x: [-r, T] \to \mathbb{R}^n$ we denote by x_t the element of C_r defined by

$$x_t(\vartheta) = x(t + \vartheta), \qquad -r \leqq \vartheta \leqq 0.$$

In this paper, the existence and uniqueness of solutions is studied for the following BVP:

$$x''(t) = f(t, Sx_t, x'(t)), \qquad t \in [0, T] \tag{1}$$

$$\begin{aligned} \alpha_0 x_0 + \alpha_1 x'(0) &= \varphi \\ \beta_0 x(T) + \beta_1 x'(T) &= \delta. \end{aligned} \tag{2}$$

Here $f: [0, T] \times B \times \mathbb{R}^n \to \mathbb{R}^n$ is a continuous function, $\varphi \in C_r$, $\varphi \in C_r$, $S: C_r \to B$ is a continuous operator and B may be equal to $\mathbb{R}^n$ or C_r or the Banach space $C([0, T], \mathbb{R}^n)$ of all continuous functions $x: [0, T] \to \mathbb{R}^n$ (with the sup-norm $\|\cdot\|_{[0,T]}$), Aldo, $\delta \in \mathbb{R}^n$ and α_i, β_i, $i = 0, 1$, are real constants satisfying

$$|\alpha_0| + |\alpha_1| > 0, \qquad |\beta_0| + |\beta_1| > 0 \tag{3}$$

and

$$\lambda = \alpha_0\beta_0 T + \alpha_0\beta_1 - \beta_0\alpha_1 \neq 0. \tag{4}$$

By $x'(0)$ and $x'(T)$ we mean $x'(0^+)$ and $x'(T^-)$, respectively.

We remark that Eq. (1) is of a general type and contains as special cases the ordinary differential equations when $Sx_t = x(t)$ and the functional

differential equations when $Sx_t = x_t$. Moreover, it contains various types of functional integrodifferential equations, when

$$Sx_t = \int_0^T h(t, s, x(t+s))\, ds, \qquad 0 \leqq t \leqq T$$

or

$$Sx_t = \int_0^T h(t, s)\, x(t+s)\, d\mu(s), \qquad -r \leqq s \leqq 0 \leqq t \leqq T,$$

where h is an appropriate function and μ is a function of bounded variation. Such operators arise when a third order BVP is transformed into a second order BVP, as the following example shows [see [1, 2]).

EXAMPLE. Consider the third order BVP

$$y'''(t) = f(t, y, y''), \qquad t \in [0, T]$$
$$y(0) = y_0$$
$$-hy'(0) + y''(0) = \gamma$$
$$ky'(T) + y''(T) = \delta.$$

Setting $x(t) = y'(t)$ we obtain

$$y(t) = y_0 + \int_0^t x(s)\, ds$$

and the above BVP is equivalent to

$$x''(t) = f\left(t, y_0 + \int_0^t x(s)\, ds, x'(t)\right)$$
$$-hx(0) + x'(0) = \gamma$$
$$kx(T) + x'(T) = \delta,$$

which is of the form (1)–(2), with

$$Sx_t = y_0 + \int_0^t x(s)\, ds$$
$$\alpha_0 = -h, \qquad \alpha_1 = 1, \qquad \beta_0 = k, \qquad \beta_1 = 1.$$

We state here some lemmas which are essential in the sequel.

LEMMA 1 [3]. *The Green's function $G(t, s)$ of the BVP*

$$\begin{aligned} x'' &= 0 \\ \alpha_0 x(0) + \alpha_1 x'(0) &= 0 \\ \beta_0 x(T) + \beta_1 x'(T) &= 0 \end{aligned} \tag{5}$$

exists if $\lambda = \alpha_0\beta_0 T + \alpha_0\beta_1 - \beta_0\alpha_1 \neq 0$, is given by

$$G(t, s) = \frac{1}{\lambda}\begin{cases} (\beta_0 t - \beta_0 T - \beta_1)(\alpha_0 s - \alpha_1), & 0 \leqq s \leqq t \leqq T \\ (\alpha_0 t - \alpha_1)(\beta_0 s - \beta_0 T - \beta_1), & 0 \leqq t \leqq s \leqq T \end{cases}$$

and the inequalities

$$\int_0^T |G(t, s)|\, ds \leqq K_1 \tag{6}$$

$$\int_0^T |G_t(t, s)|\, ds \leqq K_2 \tag{7}$$

hold, where

$$K_1 = \frac{T}{2|\lambda|}\left[\{(|\alpha_0\beta_0|\, T + 2(|\alpha_1\beta_0| + |\alpha_0\beta_1|))\, r_1 r_2 + |\alpha_0\beta_1|\, r_1^2 + |\alpha_1\beta_0|\, r_2^2\}\, T + 2|\alpha_1\beta_1|\right]$$

$$K_2 = \frac{T}{2|\lambda|}\left[|\alpha_0\beta_0|\, T + 2\max\{|\alpha_1\beta_0|, |\beta_1\alpha_0|\}\right]$$

$$r_1 = \frac{1}{2\lambda^*}\left[|\alpha_0\beta_0|\, T + 2|\alpha_0\beta_1|\right]$$

$$r_2 = \frac{1}{2\lambda^*}\left[|\alpha_0\beta_0|\, T + 2|\alpha_0\beta_0|\right]$$

$$\lambda^* = |\alpha_0\beta_0|\, T + |\alpha_0\beta_1| + |\alpha_1\beta_0|.$$

For the differential equation

$$y''(t) = \omega(t), \tag{e}$$

where $\omega\,[0, T] \to \mathbb{R}^n$ is a continuous function, using the above representation of Green's function, it is elementary to show that there exists exactly one solution y of (e) with $\alpha_0 y(0) + \alpha_1 y'(0) = \gamma$ and $\beta_0 y(T) + \beta_1 y'(T) = \delta$. We record, for reference, this result as a lemma:

LEMMA 2. *The differential equation* (e) *has exactly one solution* y *such that*

$$\alpha_0 y(0) + \alpha_1 y'(0) = \gamma$$
$$\beta_0 y(T) + \beta_1 y'(T) = \delta$$

which is given by

$$y(t) = \int_0^T G(t, s)\, \omega(s)\, ds + \frac{1}{\lambda}\,[\gamma\beta_0(T-t) + \gamma\beta_1 - \delta\alpha_1 + \delta\alpha_0 t].$$

LEMMA 3 [4, p. 113]. *Let* $B(a, r)$ *be a sphere in a complete metric space* (X, d). *Let, also,* $f: B(a, r) \to X$ *be a contraction mapping which satisfies the Lipschitz condition*

$$d(f(x_1), f(x_2)) \leqq k d(x_1, x_2)$$

for every pair of points x_1, x_2 *in* $B(a, r)$, k *being a constant such that* $0 < k < 1$. *Then if* $d(a, f(a)) \leqq r(1-k)$, *there exists a unique point* $x_0 \in B(a, r)$ *such that* $f(x_0) = x_0$.

3. EXISTENCE AND UNIQUENESS OF SOLUTIONS

In this section we establish our results on the existence and uniqueness of solutions of BVP (1)–(2) by using the well-known contraction mapping principle.

THEOREM 1. *Let* f *be continuous on the set*

$$E = [0, T] \times B \times \mathbb{R}^n$$

and satisfy on E *a Lipschitz condition of the form*

$$|f(t, x, y) - f(t, \bar{x}, \bar{y})| \leqq \theta_0 |x - \bar{x}| + \theta_1 |y - \bar{y}|, \tag{8}$$

where θ_0, θ_1 *are real constants and* $|\cdot|$ *is a convenient norm in* B. *Further, assume that the operator* S *satisfies the condition*

$$|S\varphi - S\bar{\varphi}| \leqq M \|\varphi - \bar{\varphi}\|_{[-r, 0]} \tag{9}$$

whenever $\varphi, \bar{\varphi} \in C_r$.

Then, if

$$\theta < 1, \tag{10}$$

where

$$\theta = \begin{cases} \left[\theta_0 M + \theta_1 \dfrac{K_2}{K_1}\right] \max\left\{K_1, K_2 \dfrac{|\alpha_1|}{|\alpha_0|}\right\}, & \text{if } \alpha_0 \neq 0 \\ \theta_0 K_1 M + \theta_1 K_2, & \text{if } \alpha_0 = 0 \end{cases}$$

and K_1, K_2 *are as in Lemma* 1, *for every* $(\varphi, \delta) \in C_r \times \mathbb{R}^n$ *there exists exactly one solution of BVP* (1)–(2).

Proof. Case 1. $\alpha_0 \neq 0$.

Consider the set X of all continuous functions $x: [-r, T] \to \mathbb{R}^n$ which are continuously differentiable on $[0, T]$ and such that

$$\alpha_0 x_0 + \alpha_1 x'(0) = \varphi$$

for any $x \in X$.

The formula

$$d(x, y) = \max\left\{\|x - y\|_{[0,T]}, \frac{K_1}{K_2}\|x' - y'\|_{[0,T]}, \frac{|\alpha_1|}{|\alpha_0|}|x'(0) - y'(0)|\right\}$$

defines in X a metric which makes it a complete metric space.

Let x be an arbitrary element of X. Consider the differential equation

$$y''(t) = f(t, Sx_t, x'(t)), \qquad t \in [0, T]$$

which is of the form (e) with $\omega(t) = f(t, Sx_t, x'(t))$ and, hence, by Lemma 2, there exists exactly one solution y with

$$\alpha_0 y(0) + \alpha_1 y'(0) = \gamma = \varphi(0)$$
$$\beta_0 y(T) + \beta_1 y'(T) = \delta.$$

Consider now the function

$$\tilde{x}(t) = \begin{cases} \dfrac{1}{\alpha_0}[\varphi(t) - \alpha_1 y'(0)], & t \in [-r, 0] \\ y(t), & t \in [0\ T]. \end{cases}$$

Evidently $\tilde{x} \in X$ and, thus, by the formula

$$\tilde{x} = H(x)$$

defines a mapping H of X into itself. By Lemma 2, it is easily seen that for every $x \in X$ we have

$$Hx(t)=\begin{cases}\dfrac{1}{\alpha_0}\left[\varphi(t)-\alpha_1(Hx)'(0)\right], & t\in[-r,0]\\ \displaystyle\int_0^T G(t,s)\,f(s,Sx_s,x'(s))\,ds & \\ \quad+\dfrac{1}{\lambda}\left[\gamma\beta_0(T-t)+\gamma\beta_1-\delta\alpha_1+\delta\alpha_0 t\right], & t\in[0\ T].\end{cases}\tag{11}$$

We shall prove that H is a contraction in X. Indeed, by taking into account the assumptions (8), (9), and the relations (6), (7), for every x_1, x_2 in X we have

$$\begin{aligned}
&d(Hx_1, Hx_2)\\
&\quad=\max\left\{\|Hx_1-Hx_2\|_{[0,T]},\frac{K_1}{K_2}\|(Hx_1)'-(Hx_2)'\|_{[0,T]},\right.\\
&\qquad\left.\frac{|\alpha_1|}{|\alpha_0|}|(Hx_1)'(0)-(Hx_2)'(0)|\right\}\\
&\quad\leqq\max\left\{K_1\max_{0\leqq t\leqq T}|f(t,Sx_{1t},x_1'(t)-f(t,Sx_{2t},x_2'(t))|,\right.\\
&\qquad\frac{K_1}{K_2}K_2\max_{0\leqq t\leqq T}|f(t,Sx_{1t},x_1'(t))-f(t,Sx_{2t},x_2'(t))|,\\
&\qquad\left.\frac{|\alpha_1|}{|\alpha_0|}K_2\max_{0\leqq t\leqq T}|f(t,Sx_{1t},x_1'(t))-f(t,Sx_{2t},x_2'(t))|\right\}\\
&\quad\leqq\max\left\{K_1,\frac{|\alpha_1|}{|\alpha_0|}K_2\right\}\max_{0\leqq t\leqq T}|f(t,Sx_{1t},x_1'(t))-f(t,Sx_{2t},x_2'(t))|\\
&\quad\leqq\max\left\{K_1,\frac{|\alpha_1|}{|\alpha_0|}K_2\right\}\max_{0\leqq t\leqq T}\{\theta_0M\|x_{1t}-x_{2t}\|+\theta_1\|x_1'-x_2'\|_{[0,T]}\}\\
&\quad\leqq\max\left\{K_1,\frac{|\alpha_1|}{|\alpha_0|}K_2\right\}\max_{0\leqq t\leqq T}\{\theta_0M\max(\|x_1-x_2\|_{[-r,0]},\|x_4-x_2\|_{[0,t]})\\
&\qquad+\theta_1\|x_1'-x_2'\|_{[0,T]}\}\\
&\quad\leqq\max\left\{K_1,\frac{|\alpha_1|}{|\alpha_0|}K_2\right\}\max_{0\leqq t\leqq T}\left\{\theta_0M\max\left(\frac{|\alpha_1|}{|\alpha_0|}|x_1'(0)-x_2'(0)|,\right.\right.\\
&\qquad\left.\left.\|x_1-x_2\|_{[0,t]}\right)+\theta_1\frac{K_2}{K_1}\frac{K_1}{K_2}\|x_1'-x_2'\|_{[0,T]}\right\}\\
&\quad\leqq\max\left\{K_1,\frac{|\alpha_1|}{|\alpha_0|}K_2\right\}\left[\theta_0M+\theta_1\frac{K_2}{K_1}\right]d(x_1,x_2).
\end{aligned}$$

Consequently, by (10), H is a contraction mapping and therefore there exists exactly one $x \in X$ with $Hx = x$. Obviously, x is a solution of the BVP (1)–(2). This completes the proof of the theorem, when $\alpha_0 \neq 0$.

Case 2. $\alpha_0 = 0$.

In this case we have $\alpha_1 \neq 0 \neq \beta_0$ because of (3), (4), and BVP (1)–(2) takes the form

$$x''(t) = f(t, Sx_t, x'(t)) \tag{1}$$

$$\begin{aligned} x'(0) &= \frac{\varphi}{\alpha_1} \\ \beta_0 x(T) + \beta_1 x'(T) &= \delta, \end{aligned} \tag{$\hat{2}$}$$

where φ must be constant.

Now, we consider the complete metric space $(\hat{X}, \hat{d})$, where $\hat{X}$ is the set of all continuous functions $x: [-r, T] \to \mathbb{R}^n$, which are continuously differentiable on $[0, T]$ and $x(t) = x(0)$, $t \in [-r, 0]$, endowed with the metric

$$\hat{d}(x, y) = \max \left\{ \|x - y\|_{[0,T]}, \frac{K_1}{K_2} \|x' - y'\|_{[0,T]} \right\}, \qquad x, y \text{ in } \hat{X}.$$

We define also a mapping $\hat{H}: \hat{X} \to \hat{X}$ by

$$\hat{H}x(t) = \begin{cases} \hat{H}x(0), & -r \leqq t \leqq 0 \\ \displaystyle\int_0^T G(t, s) f(s, Sx_s, x'(s))\, ds & \\ \quad - \frac{1}{\beta_0 \alpha_1} [\varphi \beta_0 (T - t) + \varphi \beta_1 - \delta \alpha_1], & 0 \leqq t \leqq T, \end{cases}$$

where

$$G(t, s) = \frac{-\alpha_1}{\lambda} \begin{cases} \beta_0 t - \beta_0 T - \beta_1, & 0 \leqq s \leqq t \leqq T \\ \beta_0 s - \beta_0 T - \beta_1, & 0 \leqq t \leqq s \leqq T. \end{cases}$$

Following the same procedure as in Case 1, we can prove that, in view of (10), $\hat{H}$ is a contraction mapping on $\hat{X}$. Thus, $\hat{H}$ has a unique fixed point x which is, obviously, the unique solution of BVP (1)–(2).

As an immediate consequence of Theorem 1, in the case when $Sx_t = x_t$, $\alpha_0 = \beta_0 = 1$, $\alpha_1 = \beta_1 = 0$ we receive the following result.

COROLLARY 1 [9, Theorem 1]. *Let f be continuous on the set*

$$E = [0, T] \times C_r \times \mathbb{R}^n$$

and satisfy on E a Lipschitz condition of the form

$$|f(t, u, v) - f(t, \bar{u}, \bar{v})| \leqq \theta_0 \|u - \bar{u}\|_{[-r,0]} + \theta_1 |v - \bar{v}|$$

with Lipschitz constants θ_0 and θ_1 such that

$$\theta_0 \frac{T^2}{8} + \theta_1 \frac{T}{2} < 1.$$

Then, for every $(\varphi, \eta) \in C_r \times \mathbb{R}^n$ there exists exactly one solution of the BVP

$$x''(t) = f(t, x_t, x'(t))$$
$$x_0 = \varphi, \qquad x(T) = \eta.$$

If in the above corollary $S\varphi = \varphi(0)$ then we have the results of Hartman [5].

In the case when $Sx_t = \int_0^T h(t, s, x_s)\, ds$, we have the following result, which generalizes the results of Morchalo [7] and Pachpatte [10] (which concern ordinary differential equations and $\alpha_0 = \beta_0 = 1$, $\alpha_1 = \beta_1 = 0$).

COROLLARY 2. *Let the functions f and h satisfy the conditions*

$$|f(t, x_1, y_1) - f(t, x_2, y_2)| \leqq \theta_0 |x_1 - x_2| + \theta_1 |y_1 - y_2|$$
$$|h(t, s, \varphi_1) - h(t, s, \varphi_2)| \leqq M \|\varphi_1 - \varphi_2\|_{[-r,0]}$$

for all $(t, s, \varphi_i) \in [0, T] \times [0, T] \times C_r$, $(t, x_i, y_i) \in [0, T] \times \mathbb{R}^n \times \mathbb{R}^n$, $i = 1, 2$, where θ_0, θ_1, M are nonnegative constants such that

$$\hat{\theta} < 1,$$

where

$$\hat{\theta} = \begin{cases} \left[\theta_0 M + \theta_1 \dfrac{K_2}{K_1}\right] \max\left\{K_1, K_2 \dfrac{|\alpha_1|}{|\alpha_0|}\right\}, & \text{if } \alpha_0 \neq 0 \\ \theta_0 K_1 M + \theta_1 K_2, & \text{if } \alpha_0 = 0 \end{cases}$$

Then, the BVP

$$x''(t) = f\left(t, \int_0^t h(t, s, x_s)\, ds, x'(t)\right), \qquad t \in [0, T]$$

$$\alpha_0 x_0 + \alpha_1 x'(0) = \varphi$$

$$\beta_0 x(T) + \beta_1 x'(T) = \delta$$

has exactly one solution.

The problem of formulating conditions under which a BVP has a priori bounded solutions is a very interesting problem in the theory of boundary value problems. (See Hartman [5].)

In the next theorem we obtain conditions such that the BVP has a priori bounded solutions. This result is achieved by a well-known variation of the contraction principle given in Lemma 3.

Before stating our Theorem 2, we establish some notations which simplify the proof below.

Thus we set

$$F = \begin{cases} \|f(t, S\Phi_t, 0)\|_{[0,T]}, & \text{if} \quad \alpha_0 \neq 0 \\ \|f(t, S0, 0)\|_{[0,T]}, & \text{if} \quad \alpha_0 = 0, \end{cases}$$

where

$$\Phi(t) = \begin{cases} \left\{ \begin{array}{ll} \dfrac{\varphi(t)}{\alpha_0}, & -r \leqq t \leqq 0 \\ \dfrac{\varphi(0)}{\alpha_0}, & 0 \leqq t \leqq T \end{array} \right\}, & \text{if} \quad \alpha_0 \neq 0 \\ 0, \quad -r \leqq t \leqq T, & \text{if} \quad \alpha_0 = 0. \end{cases}$$

Also,

$$L = \begin{cases} \left\| l(t) - \dfrac{\varphi(0)}{\alpha_0} \right\|_{[0,T]}, & \text{if} \quad \alpha_0 \neq 0 \\ \|l(t)\|_{[0,T]}, & \text{if} \quad \alpha_0 = 0 \end{cases}$$

$$L' = \left| \frac{\varphi(0)\, \beta_0 + \delta\alpha_0}{\lambda} \right|,$$

$$\theta = \begin{cases} \left(\theta_0 M + \theta_1 \dfrac{K_2}{K_1}\right) \max\left\{K_1, \dfrac{|\alpha_1|}{|\alpha_0|} K_2\right\}, & \text{if} \quad \alpha_0 \neq 0 \\ \theta_0 K_1 M + \theta_1 K_2, & \text{if} \quad \alpha_0 = 0 \end{cases}$$

and

$$K=\begin{cases}\max\left\{K_1F+L,\,(K_2F+L')\max\left\{\dfrac{K_1}{K_2},\dfrac{|\alpha_1|}{|\alpha_0|}\right\}\right\} & \text{if}\quad \alpha_0\neq 0\\ \max\left\{K_1F+L,\,(K_2F+L')\dfrac{K_1}{K_2}\right\} & \text{if}\quad \alpha_0=0.\end{cases}$$

THEOREM 2. *If conditions* (8), (9), *and* (10) *of Theorem* 1 *are satisfied, then for every* r *with*

$$r\geqq\frac{K}{1-\theta},\tag{12}$$

the BVP (1)–(2) *has a unique solution* x *such that*

$$\|x\|_{[0,T]}\leqq r+q$$

and

$$\|x'\|_{[0,T]}\leqq\frac{K_1}{K_2}r,$$

where

$$q=\begin{cases}\left|\dfrac{\varphi(0)}{\alpha_0}\right|, & \text{if}\quad \alpha_0\neq 0\\ 0, & \text{if}\quad \alpha_0=0.\end{cases}$$

Proof. Case 1. $\alpha_0\neq 0$.

Consider the metric space (X, d) as in the proof of Theorem 1. Then the function Φ defined by

$$\Phi(t)=\begin{cases}\dfrac{\varphi(t)}{\alpha_0}, & -r\leqq t\leqq 0\\ \dfrac{\varphi(0)}{\alpha_0}, & 0\leqq t\leqq T\end{cases}$$

is obviously an element of X. Let $B(\Phi, r)$ be a sphere in the metric space X, with radius r such that

$$r\geqq\frac{K}{1-\theta}.$$

It is clear, from the proof of Theorem 1, that the mapping H: $B(\Phi, r)\to X$,

defined by (11) is a contraction. So, following Lemma 3, we must check that

$$d(\Phi, H\Phi) \leqq r(1-\theta), \tag{13}$$

where, by Theorem 1, $\theta = \max\{K_1, (|\alpha_1|/|\alpha_0|)\, K_2\}[\theta_0 M + \theta_1(K_2/K_1)] < 1$ is the contraction constant of H.

To this end we observe that

$$\|H\Phi - \Phi\|_{[0,T]} \leqq K_1 F + L \tag{14}$$

and

$$\|(H\Phi)' - \Phi'\|_{[0,T]} \leqq K_2 F + L'. \tag{15}$$

In view of (14), (15), condition (13) is satisfied provided that

$$r \geqq \frac{1}{1-\theta} \max\left\{K_1 F + K, \left(\max\left\{\frac{K_1}{K_2}, \frac{|\alpha_1|}{|\alpha_0|}\right\}\right)(K_2 F + L')\right\} = \frac{K}{1-\theta}$$

because of (12).

Therefore, by Lemma 3, there exists a unique fixed point $x \in B(\Phi, r)$ of the mapping H or, equivalently, there exists a unique solution x of the BVP such that

$$d(x, \Phi) \leqq r. \tag{16}$$

But it is easily seen that from (16) we obtain

$$\|x\|_{[0,T]} \leqq r + \left|\frac{\varphi(0)}{\alpha_0}\right| = r + \left|\frac{\gamma}{\alpha_0}\right|$$

and

$$\|x'\|_{[0,T]} \leqq \frac{K_2}{K_1} r.$$

Case 2. $\alpha_0 = 0$.

As in Case 2 of Theorem 1, we consider the metric space $(\hat{X}, \hat{d})$. We take also $\Phi = 0$ and the sphere $B(0, r)$ in $\hat{X}$, where r satisfies (12).

Consider also the mapping $\hat{H}: B(0, r) \to \hat{X}$, as in Case 2 in Theorem 1 and, by the same way, we can show that $\hat{H}$ is a contraction. Thus, there exists a unique fixed point $x \in B(0, r)$ of $\hat{H}$, or, in other words, there exists a unique solution x of BVP (1)–(2) which satisfies

$$\|x\|_{[0,T]} \leqq r$$

$$\|x'\|_{[0,T]} \leqq \frac{K_1}{K_2} r.$$

4. Continuous Dependence

The results of the previous paragraphs remain valid if instead of equation (1) we have the equation

$$x''(t) = f(t, Sx_t, x'(t), u), \tag{1_u}$$

where u is a real parameter.

Indeed, it suffices to consider the operator $H_u: X \to X$ defined by the formula

$$H_u x(t) = \begin{cases} \dfrac{1}{\alpha_0}[\varphi(t) - \alpha_1 x'(0)], & -r \leqq t \leqq 0 \\ \displaystyle\int_0^T G(t, s) f(s, Sx_s, x'(s), u)\, ds & \\ \quad + \dfrac{1}{\lambda}[\gamma\beta_0(T-t) + \gamma\beta_1 - \delta\alpha_1 + \delta\alpha_0 t], & 0 \leqq t \leqq T, \end{cases}$$

if $\alpha_0 \neq 0$ and the operator $\hat{H}_u: \hat{X} \to \hat{X}$, defined by the formula

$$\hat{H}_u x(t) = \begin{cases} \hat{H}_u x(0), & -r \leqq t \leqq 0 \\ \displaystyle\int_0^T G(t, s) f(s, Sx_s, x'(s); u)\, ds & \\ \quad - \dfrac{1}{\beta_0\alpha_1}[\varphi\beta_0(T-t) + \varphi\beta_1 - \delta\alpha_1], & 0 \leqq t \leqq T \end{cases}$$

if $\alpha_0 = 0$. It is clear that the proofs of Theorems 1 and 2 are valid without changes for the boundary value problems (1_u)–(2) and (1_u)–$(\hat{2})$, by setting H_u and $\hat{H}_u$ in place of H and $\hat{H}$, respectively.

In this section we deal with the problem of dependence of solutions of BVP (1_u)–(2) or of BVP (1_u)–$(\hat{2})$, on the parameter u. More precisely, we will show the following.

Theorem 3. *Assume that the conditions* (8), (9), *and* (10) *of Theorem* 1 (*and, moreover* (12) *of Theorem* 2, *respectively*) *are valid. Also, suppose that there exists a constant* N *such that*

$$|f(t, x, y, u_1) - f(t, x, y, u_2)| \leqq N|u_1 - u_2|, \tag{17}$$

for every $(t, x, y) \in E$ *and* u_1, u_2 *in* $\mathbb{R}$.

Then the solution of the boundary value problem (1_u)–(2), $u \in \mathbb{R}$, *obtained by Theorem* 1 (*obtained by Theorem* 2, *respectively*) *depends continuously on* u.

Proof. *Case* 1. $\alpha_0 \neq 0$.

As in Theorem 1 (Theorem 2, respectively) for every x_1, x_2 in X and $u \in \mathbb{R}$ we have

$$d(H_u x_1, H_u x_2) \leqq \theta d(x_1, x_2). \tag{18}$$

Also, for every $x \in X$ and u_1, u_2 in $\mathbb{R}$ from (17) we obtain

$$\begin{aligned} d(H_{u_1}x, H_{u_2}x) &= \max\left\{ \|H_{u_1}x - H_{u_2}x\|_{[0,T]}, \frac{K_1}{K_2} \|(H_{u_1}x)' \right. \\ &\quad \left. - (H_{u_2}x)'\|_{[0,T]}, \frac{|\alpha_1|}{|\alpha_0|} |(H_{u_1}x)'(0) - (H_{u_2}x)'(0)| \right\} \\ &= \max\left\{ K_1 \max_{0 \leqq t \leqq T} \Delta(t, x, u_1, u_2), \frac{K_1}{K_2} K_2 \max_{0 \leqq t \leqq T} \Delta(t, x, u_1, u_2), \right. \\ &\quad \left. \left|\frac{\alpha_1}{\alpha_0}\right| K_2 \max_{0 \leqq t \leqq T} \Delta(t, x, u_1, u_2) \right\}, \end{aligned}$$

where

$$\Delta(t, x, u_1, u_2) = |f(t, Sx_t, x'(t), u_1) - f(t, Sx_t, x'(t), u_2)| \leqq N|u_1 - u_2|,$$

because of (17).

Hence

$$d(H_{u_1}x, H_{u_2}x) \leqq \max\left\{ K_1, \frac{|\alpha_1|}{|\alpha_0|} K_2 \right\} N|u_1 - u_2|$$

or

$$d(H_{u_1}x, H_{u_2}x) \leqq N_1|u_1 - u_2|, \tag{19}$$

where $N_1 = N \max\{K_1, (|\alpha_1|/|\alpha_0|) K_2\}$.

But Theorem 1 (Theorem 2, respectively) guarantees that for every $u \in \mathbb{R}$ there exists a function $x_u \in X$ such that

$$x_u = H_u x_u.$$

Thus from (18) and (19) we have

$$\begin{aligned} d(x_{u_1}, x_{u_2}) &= d(H_{u_1}x_{u_1}, H_{u_2}x_{u_2}) \\ &\leqq d(H_{u_1}x_{u_1}, H_{u_1}x_{u_2}) + d(H_{u_1}x_{u_2}, H_{u_2}x_{u_2}) \\ &\leqq \theta d(x_{u_1}, x_{u_2}) + N_1|u_1 - u_2|. \end{aligned}$$

Hence,

$$d(x_{u_1}, x_{u_2}) \leqq (1-\theta)^{-1} N_1 |u_1 - u_2|$$

which proves our Theorem when $\alpha_0 \neq 0$.

Case 2. $\alpha_0 \neq 0$.

In this case, following entirely similar reasoning as in Case 1, we obtain

$$\hat{d}(x_{u_1}, x_{u_2}) \leqq (1-\theta)^{-1} \hat{N}_1 |u_1 - u_2|,$$

where $\hat{N}_1 = NK_1$.

Obviously, here we consider the metric space $(\hat{X}, \hat{d})$ and the operator $\hat{H}_u$.

REFERENCES

1. A. R. AFTABIZADEH AND J. WIENER, Existence and uniqueness theorems for third order boundary value problems, *Rend. Sem. Mat. Univ. Padova* **75** (1986), 129–141.
2. A. R. AFTABIZADEH AND S. LEELA, Existence results for Boundary value problem of Integrodifferential equations, *in* "Colloquia Mathematica Societatis Janos Bolyai," North-Holland, Amsterdam, 1984.
3. R. P. AGARWAL, Boundary value problems for higher order differential equations, World Scientific, Singapore/Philadelphia, 1986.
4. E. T. COPSON, "Metric Spaces," Cambridge Univ. Press, Cambridge, 1972.
5. P. HARTMAN, "Ordinary Differential Equations," Wiley–Interscience, New York, 1964.
6. G. S. LADDE AND B. G. PACHPATTE, Existence theorems for a class of functional differential systems, *J. Math. Anal. Appl.* **90** (1982), 381–392.
7. J. MORCHALO, On two point boundary value problem for an integrodifferential equation of second order, *Fasc. Math.* **9** (1975), 51–66.
8. T. S. NIKOLOVA AND D. D. BAINOV, Application of spline-functions for the construction of an approximate solution of boundary value problems for a class of functional differential equations, *Yokohama Math. J.* **29** (1981), 107–122.
9. S. K. NTOUYAS, A boundary value problem for second order functional differential equations, *Hiroshima Math. J.* **12** (1982), 453–468.
10. B. G. PACHPATTE, Boundary value problems for nonlinear second order integrodifferential equations, *An. Stint. Univ. "Al. I. Cuza" Iasi, Sect. Ia Mat. (N.S.)* **31** (1985), 65–68.